# Bayes Learning — MCQs

**1. Which theorem forms the mathematical foundation of Bayes Learning?**

A. Central Limit Theorem
**B. Bayes' Theorem**
C. Maximum Likelihood Theorem
D. Markov Theorem

**2. Which of the following correctly represents Bayes' theorem?**

A. $P(A \mid B) = \frac{P(A)P(B)}{P(B \mid A)}$

**Answer : B.** $P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$

C. $P(A \mid B) = P(A) + P(B)$

D. $P(A \mid B) = P(A) - P(B)$

**3. In Bayes' theorem, $P(A)$ is referred to as:**

A. Likelihood
B. Posterior
**C. Prior**
D. Evidence

**4. What does $P(A \mid B)$ represent?**

A. Probability of B given A
**B. Probability of A given B**
C. Probability of A and B independently
D. Probability of B occurring

**5. What does $P(B \mid A)$ represent in Bayes' theorem?**

A. Prior probability
B. Posterior probability
**C. Likelihood**
D. Marginal probability

**6. A classifier chooses the class having the highest posterior probability. Which principle is it using?**

A. Maximum Likelihood Estimation
**B. Maximum A Posteriori estimation**

C. Gradient Descent
D. Principal Component Analysis

**7. Suppose $P(A) = 0.4$, $P(B \mid A) = 0.5$, and $P(B) = 0.2$. What is $P(A \mid B)$?**

A. 0.10
B. 0.25
C. 0.50
**D. 0.80 Answer getting is 1 this is the one close to. It**

**8. A disease occurs in 1% of a population. A diagnostic test has a sensitivity of 90% and a false-positive rate of 5%. If a randomly selected person tests positive, which factor is most important when determining the probability that the person actually has the disease?**

A. Only the sensitivity
B. Only the false-positive rate
**C. The disease prevalence as well as the test characteristics**
D. Only the number of people tested

**9. Why can a highly accurate diagnostic test still have a relatively low positive predictive value?**

**A. The prior probability of the disease may be very low**
B. Bayes' theorem ignores false positives
C. Sensitivity is always lower than specificity
D. Accuracy and probability are unrelated

**10. Which of the following best describes Bayesian learning?**

A. Learning only from observed data
**B. Updating prior beliefs using observed evidence**
C. Ignoring prior knowledge completely
D. Selecting parameters randomly

**11. In Bayesian learning, what happens to the posterior probability when new evidence becomes available?**

A. It remains equal to the prior probability
**B. It is updated based on the evidence**
C. It always becomes zero
D. It always becomes one

**12. Which statement correctly distinguishes prior and posterior probability?**

A. Prior is after observing evidence; posterior is before evidence
B. Prior and posterior are always identical
**C. Prior represents belief before evidence; posterior represents updated belief after evidence**
D. Posterior does not depend on evidence

**13. In Naive Bayes classification, the "naive" assumption is that:**

A. All classes have equal probability
**B. Features are conditionally independent given the class**
C. Features are completely independent of the class
D. All features have the same distribution

**14. A Naive Bayes classifier receives three features for a sample. Under its standard assumption, it treats these features as:**

**A. Conditionally independent given the class**
B. Completely dependent on each other
C. Identical
D. Irrelevant to the class

**15. Which of the following is a potential problem with Naive Bayes?**

A. It cannot calculate probabilities
**B. Its conditional-independence assumption may not hold in real-world data**
C. It requires neural networks
D. It cannot perform classification

**16. Suppose two classes have identical likelihoods for a particular observation, but their prior probabilities are different. Which class will Bayesian classification favor?**

A. The class with the smaller prior
**B. The class with the larger prior**
C. Both classes equally
D. The class with more features

**17. If $P(A \mid B) \neq P(B \mid A)$, which statement is correct?**

A. This violates Bayes' theorem
**B. Conditional probabilities are generally not symmetric**
C. One of the probabilities must be zero
D. A and B must be independent

**18. If two events A and B are independent, which relationship holds?**

**Answer** :A. $P(A \mid B) = P(A)$
B. $P(A \mid B) = P(B)$
C. $P(A \mid B) = 1 - P(A)$
D. $P(A \mid B) = 0$

**19. If the prior probability of a class is extremely small, can its posterior probability still become large?**

A. No, the prior completely determines the posterior
**B. Yes, sufficiently strong evidence can overcome a low prior**
C. No, posterior probability cannot exceed the prior
D. Only if the classes are equally likely

**20. Two models produce the following probabilities for a hypothesis $H$:**

- Model A: Prior = 0.8, Likelihood = 0.2
- Model B: Prior = 0.2, Likelihood = 0.8

Which statement is most appropriate?

A. Model A must always have the larger posterior
B. Model B must always have the larger posterior
**C. The posterior cannot be determined from these values alone without considering the evidence probability**
D. Both models must have the same posterior